CSEC Mathematics

January 2019 – Paper 2

Solutions

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SECTION I

Answer ALL questions.

All working must be clearly shown.

1. (a) Evaluate

(i) $3.8 \times 10^2 + 1.7 \times 10^3$, giving your answer in standard form [2]

$$3.8 \times 10^2 + 1.7 \times 10^3 = (3.8 \times 10^2) + (1.7 \times 10^3)$$
$$= 380 + 1700$$
$$= 2080$$
$$= 2.08 \times 10^3 \quad \text{[in standard form]}$$

(ii) $\dfrac{\frac{1}{2} \times \frac{3}{5}}{3\frac{1}{2}}$, giving your answer as a fraction in its lowest terms. [2]

$$\text{Numerator} = \frac{1}{2} \times \frac{3}{5}$$
$$= \frac{3}{10}$$

$$\text{Denominator} = 3\frac{1}{2}$$
$$= \frac{7}{2}$$

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$\therefore$ Numerator $\div$ Denominator $= \frac{3}{10} \div \frac{7}{2}$

$= \frac{3}{10} \times \frac{2}{7}$

$= \frac{3}{35}$

(b) Express the number 6 as a binary number. [1]

2	6	
2	3	r 0
2	1	r 1
	0	r 1

$\therefore 6_{10} = 110_2$

(c) John bought a car for \$65 000. If the value of the car depreciates by 8% each year, how much will the car be worth at the end of 2 years? [2]

Initial cost of the car = \$65 000.

Percentage depreciation = 8%

Value of car after depreciation = $(100 - 8)\%$

$= 92\%$

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Value of the car after 1 year $= 92\%\ of\ \$65\ 000$

$$= \frac{92}{100} \times 65\ 000$$

$$= \$59\ 800$$

Value of the car after 2 years $= 92\%\ of\ \$59\ 800$

$$= \frac{92}{100} \times 59\ 800$$

$$= \$55\ 016$$

(d) The table below shows the results obtained by a student in her CSEC Mathematics examination. The maximum mark for each paper is given in the third column of the table.

Paper	**Percentage Obtained**	**Maximum Mark for Paper**
01	55	30
02	60	50
03	80	20
Total		**100**

Determine, as a percentage, the student's final mark for the Mathematics examination. [2]

For Paper 01: Mark $= \frac{55}{100} \times 30$

$$= 16.5$$

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For Paper 02: Mark $= \frac{60}{100} \times 50$

$= 30$

For Paper 03: Mark $= \frac{80}{100} \times 20$

$= 16$

Total Percentage Obtained $= \frac{16.5+30+16}{100} \times 100$

$= 62.5\%$

Total: 9 marks

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2. (a) (i) Make x the subject of the formula

$$y = \frac{x}{5} + 3p$$ [2]

$$y = \frac{x}{5} + 3p$$

$$\frac{x}{5} + 3p = y$$

$$\frac{x}{5} = y - 3p$$

$$x = 5(y - 3p)$$

(ii) Solve the following equation by factorization.

$$2x^2 - 9x = 0$$ [2]

$$2x^2 - 9x = 0$$

$$x(2x - 9) = 0$$

Either

$x = 0$ or $2x - 9 = 0$

$$2x = 9$$

$$x = \frac{9}{2}$$

$\therefore x = 0$ or $x = \frac{9}{2}$

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(b) A farmer wishes to enclose a rectangular plot with a wire fence. The width of the plot is 3 metres less than its length, l.

Given that the area enclosed by the fence is 378 square metres, show that

$$l^2 - 3l - 378 = 0 \qquad [2]$$

Since the width is 3 metres less than the length, then the width is $(l - 3)$.

We are given that the area is 378 m^2.

So, we have,

$$\text{Area} = \text{length} \times \text{width}$$

$$378 = l \times (l - 3)$$

$$378 = l^2 - 3l$$

$$l^2 - 3l - 378 = 0$$

Q.E.D.

(c) The force, F, applied to an object is directly proportional to the extension, e, produced by that object.

(i) Represent this information as an equation in terms of F, e and an appropriate constant, k. [1]

$$\text{Force} \propto \text{extension}$$

$$F \propto e$$

$$\therefore F = ke$$

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(ii) The incomplete table below shows corresponding values of F and e.

$\boldsymbol{F}$	8	25	60	$\boldsymbol{y}$
$\boldsymbol{e}$	0.2	$\boldsymbol{x}$	1.5	3.2

Using the equation obtained in (c)(i), or otherwise, determine the value of x and y. [2]

From the table, $F = 0.8$ when $e = 0.2$. So, we have,

$0.8 = k(0.2)$

$k = \frac{0.8}{0.2}$

$k = 40$

$\therefore F = 40e$

When $F = 25, e = x$. So, we have,

$25 = 40x$

$x = \frac{25}{40}$

$x = \frac{5}{8}$

When $F = y, e = 3.2$.

So, we have,

$y = 40(3.2)$

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$y = 128$

$\therefore x = 0.625$ and $y = 128$

Total: 9 marks

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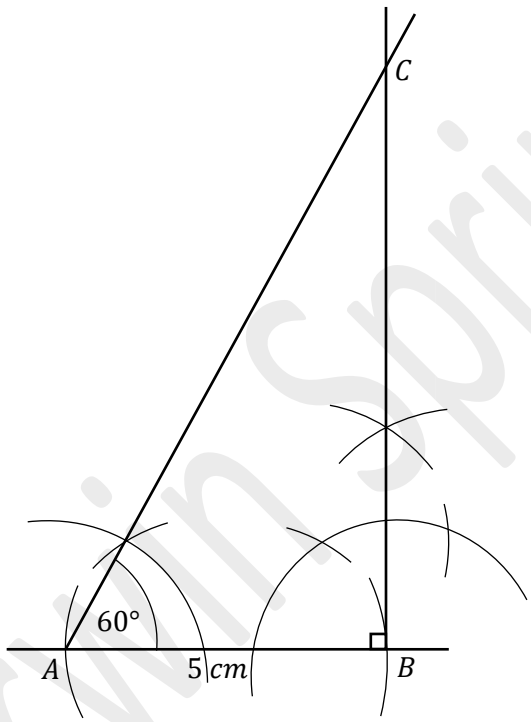
3. (a) Using a ruler, a pencil and a pair of compasses, construct the **right-angled** triangle ABC, such that $AB = 5$ cm, $\angle ABC = 90°$ and $\angle BAC = 60°$. [4]

$AB = 5\ cm$, $\angle ABC = 90°$ and $\angle BAC = 60°$



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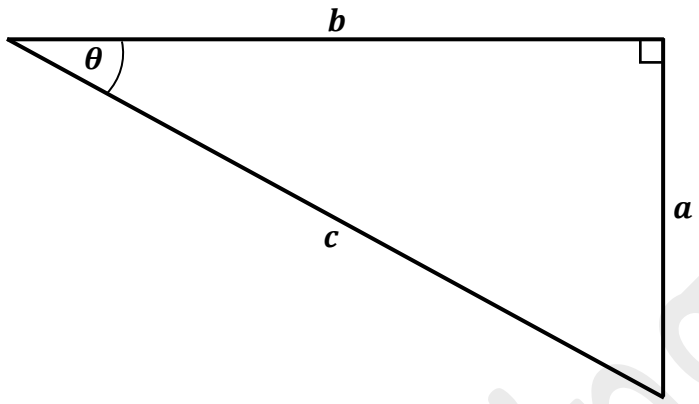
(b) The diagram below shows a right-angled triangle with sides a units, b units and c units.



(i) Using the diagram,

(a) express c in terms of a and b [1]

Using Pythagoras' Theorem,

$c^2 = a^2 + b^2$

$c = \sqrt{a^2 + b^2}$

(b) write, in terms of a, b and c, an expression for $\sin\theta + \cos\theta$. [2]

$\sin\theta = \frac{opp}{adj}$

$= \frac{a}{c}$

$\cos\theta = \frac{adj}{hyp}$

$= \frac{b}{c}$

$\therefore \sin\theta + \cos\theta = \frac{a}{c} + \frac{b}{c}$

$= \frac{a+b}{c}$

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(ii) Using the results from (i)(a) and (b) **on page 12**, show that

$(\sin\theta)^2 + (\cos\theta)^2 = 1.$ [2]

$$(\sin\theta)^2 + (\cos\theta)^2 = \left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2$$

$$= \frac{a^2}{c^2} + \frac{b^2}{c^2}$$

$$= \frac{a^2+b^2}{c^2}$$

Since $c^2 = a^2 + b^2$, then we have

$$(\sin\theta)^2 + (\cos\theta)^2 = \frac{c^2}{c^2}$$

$$\therefore (\sin\theta)^2 + (\cos\theta)^2 = 1$$

Total: 9 marks

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4. (a) Given the function $h(x) = \frac{2x+3}{5-x}$, determine

(i) the value of x for which the function is undefined [1]

The function is undefined when the denominator equals zero. So, we have,

$x - 5 = 0$

$x = 5$

(ii) an expression for $h^{-1}(x)$. [3]

$h(x) = \frac{2x+3}{5-x}$

Let $y = h(x)$.

$y = \frac{2x+3}{5-x}$

Interchange variables x and y.

$x = \frac{2y+3}{5-y}$

Make y the subject of the formula.

$x(5 - y) = 2y + 3$

$5x - xy = 2y + 3$

$2y + xy = 5x - 3$

$y(2 + x) = 5x - 3$

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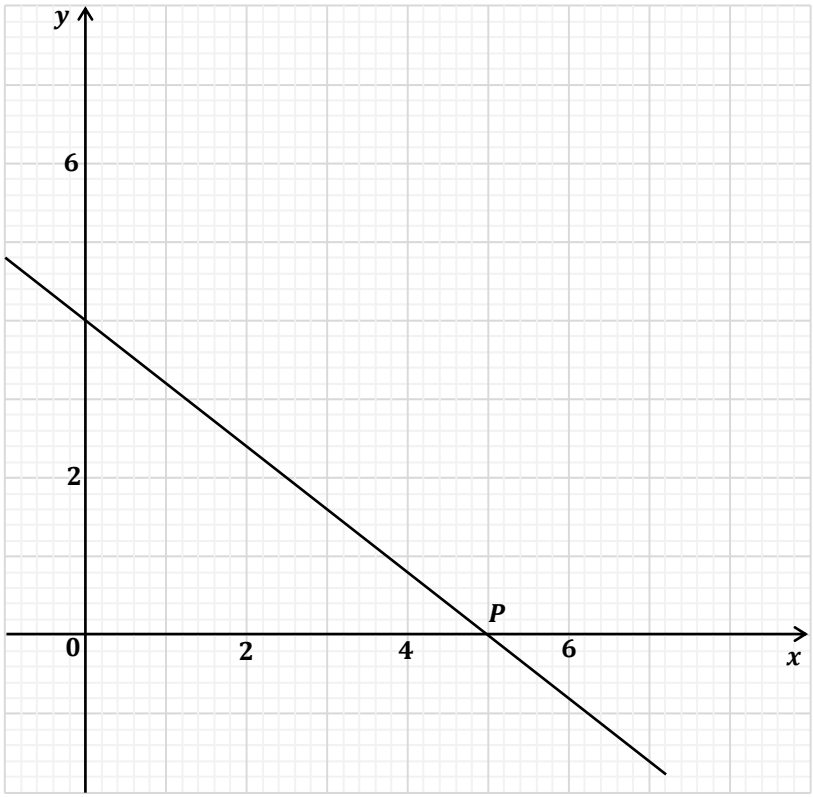
$y = \frac{5x-3}{2+x}$

$\therefore h^{-1}(x) = \frac{5x-3}{2+x}$

(b) The graph below shows a straight line intersecting the x and y axes.



Using the graph, determine the

(i) gradient of the line [2]

Two points on the line are $(0, 4)$ and $(5, 0)$.

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{0-4}{5-0}$

$= -\frac{4}{5}$

(ii) equation of the line [1]

Since the graph cuts the y-axis at $(0, 4)$, then the y-intercept $c = 4$.

Substituting $m = -\frac{4}{5}$ and $c = 4$ into the equation of the line gives:

$y = mx + c$

$y = -\frac{4}{5}x + 4$

(iii) equation of the perpendicular line that passes through P [2]

The gradient of the perpendicular line $= \frac{-1}{-\frac{4}{5}}$

$= \frac{5}{4}$

The equation of the perpendicular line is:

$y - 0 = \frac{5}{4}(x - 5)$

$y = \frac{5}{4}x - \frac{25}{4}$ or $4y = 5x - 25$

Total: 9 marks

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5. (a) A survey was conducted among 48 persons to find out what mobile network they used. The table below shows the results of the survey.

Mobile Network	WireTech	DigiLec	O-Fone	Other
Number of Persons	20	12	10	6

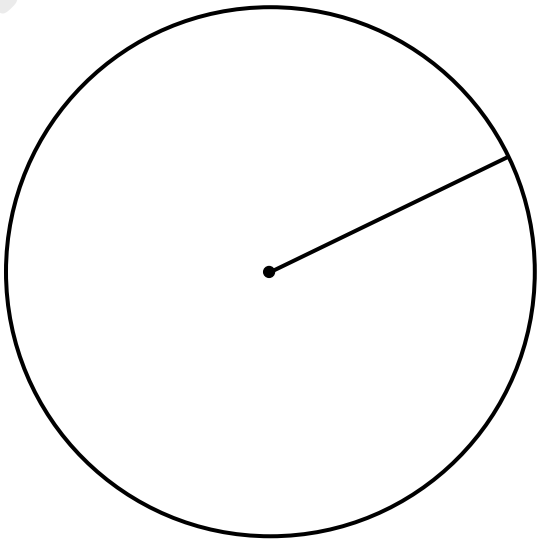
(i) If this information is to represented on a pie chart, what is the angle for the sector that will represent O-Fone? [2]

$$\text{Angle of the sector to represent O-Fone} = \frac{Number\ of\ persons\ who\ use\ O{-}Fone}{Total\ number\ of\ persons\ in\ the\ survey} \times 360°$$

$$= \frac{10}{48} \times 360°$$

$$= 75°$$

(ii) Using the circle below, with radius shown, represent the information in the table above on a clearly labelled pie chart. [3]



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Angle of the sector to represent WireTech $= \frac{Number\ of\ persons\ who\ use\ \text{WireTech}}{Total\ number\ of\ persons\ in\ the\ survey} \times 360°$

$= \frac{20}{48} \times 360°$

$= 150°$

Angle of the sector to represent DigiLec $= \frac{Number\ of\ persons\ who\ use\ \text{DigiLec}}{Total\ number\ of\ persons\ in\ the\ survey} \times 360°$

$= \frac{12}{48} \times 360°$

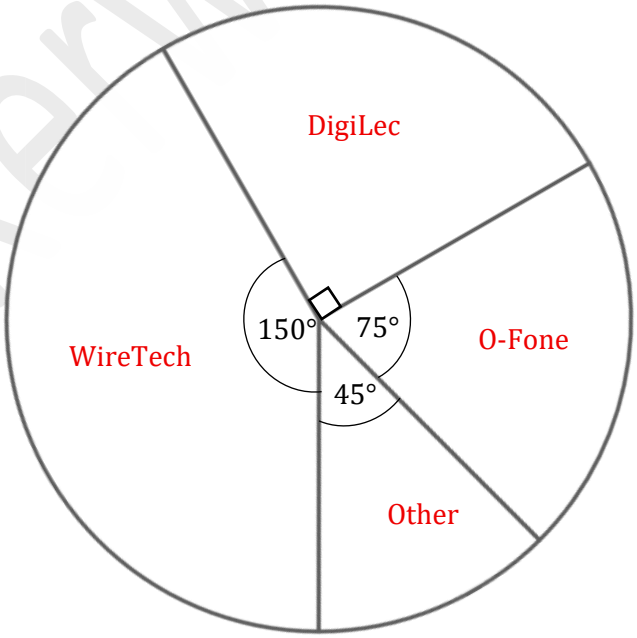
$= 90°$

Angle of the sector to represent Other $= \frac{Number\ of\ persons\ who\ use\ \text{Other}}{Total\ number\ of\ persons\ in\ the\ survey} \times 360°$

$= \frac{6}{48} \times 360°$

$= 45°$

Title: Pie Chart showing the mobile network used by 48 persons.



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(b) The incomplete table below shows the results obtained by 200 boys and 250 girls on a Spanish examination.

Grade	Boys	Girls
I		62
II		75
III		90
IV	30	
V	8	
Total	**200**	**250**
Standard deviation	8.2	6.3

(i) A girl is chosen at random. Determine the probability that she achieve a Grade I. [1]

$$P(girl\ achieves\ Grade\ 1) = \frac{Number\ of\ girls\ who\ achieved\ Grade\ 1}{Total\ number\ of\ girls}$$

$$= \frac{62}{250}$$

$$= \frac{31}{125}$$

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(ii) What percentage of the boys who took the exam achieved Grades I to III?

[2]

Number of boys who achieved Grades I to III $= 200 - (30 + 8)$

$= 162$

Percentage $= \frac{162}{200} \times 100$

$= 81\%$

(iii) Considering the standard deviations in the table **on page 18**, compare the performance of the boys and the girls. [1]

The standard deviation of the boys' performances is greater than that of the girls. Since standard deviation measures the spread of the data, this indicates that the boys' marks are more widely dispersed around the mean, while the girls' marks are more consistent and closer to their mean.

Total: 9 marks

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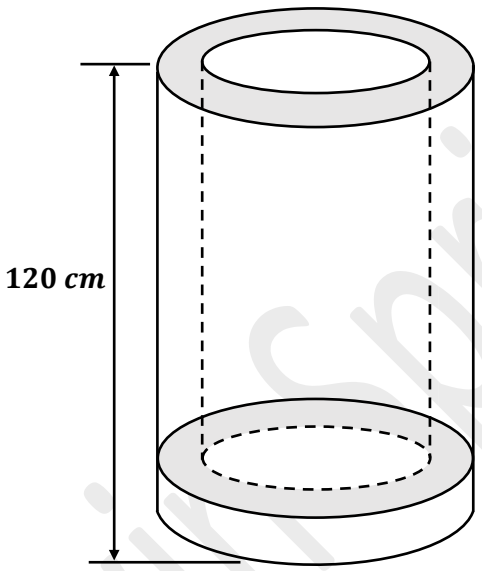
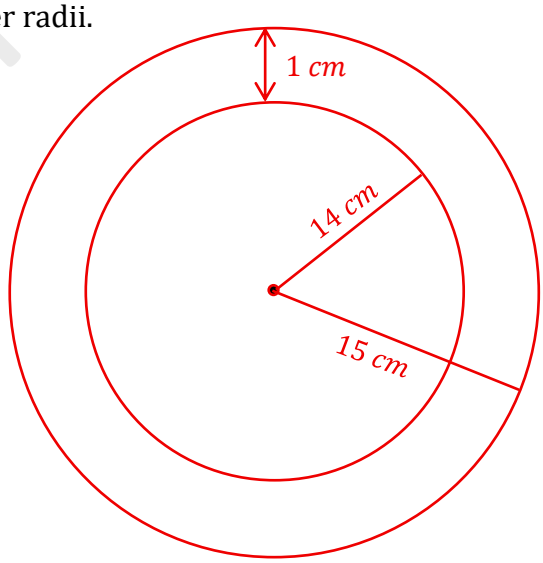
6. The diagram below, **not drawn to scale**, shows an open cylindrical container made of metal with a circular base and **uniform thickness throughout**. The length of the container, from the top to outer bottom, is 120 cm and the inner and outer radii are 14 cm and 15 cm respectively.

Take π to be $\frac{22}{7}$.



(a) Draw a cross-sectional view of the container showing the measurements of the inner and outer radii. [2]



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(b) Show that the capacity of the container is 73 304 cm^3. [2]

Volume $= \pi r^2 h$

$= \frac{22}{7} \times (14)^2 \times 119$

$= 73\ 304\ cm^3$

Hence, the capacity of the container is 73 304 cm^3.

(c) Determine the volume of the material used to make the container. [3]

External volume of the cylinder $= \pi r^2 h$

$= \frac{22}{7} \times (15)^2 \times 120$

$= 84\ 857.14\ cm^3$

Volume of the material used

= External volume of the container − Internal volume of the container

$= 84\ 857.14 - 73\ 304$

$= 11\ 553.14\ cm^3$ (to 2 decimal places)

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(d) Given that the density of the material used to make the container is 2.2 g/cm^3,

determine the mass, in kg, of the empty container.

$\left(\textbf{density} = \frac{\textbf{mass}}{\textbf{volume}}\right)$ [2]

$$\text{Density} = \frac{Mass}{Volume}$$

$$\therefore \text{Mass} = \text{Density} \times \text{Volume}$$

$$= 2.2 \times 11\,553.14$$

$$= 25\,416.91\ g$$

Now,

$$1000\ g = 1\ kg$$

$$1\ g = \frac{1}{1000}$$

$$25\,416.91\ g = \frac{1}{1000} \times 25\,416.91$$

$$= 25.4\ kg \quad \text{(to 2 decimal places)}$$

Total: 9 marks

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7. A sequence of figures is made from joining polygons with sides of unit length. The first three figures in the sequence are shown below.

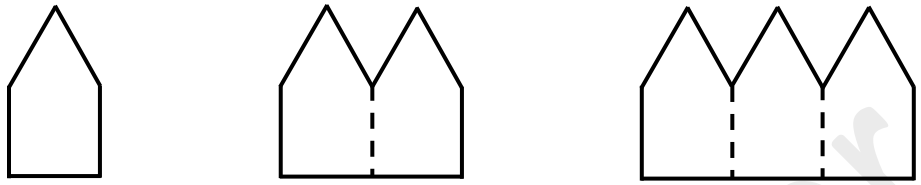
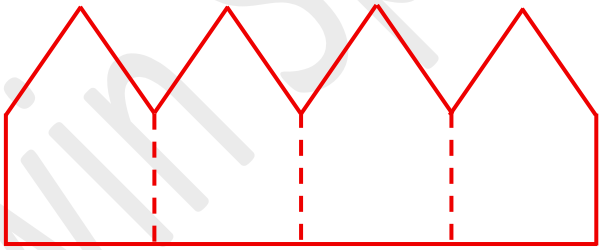


Figure 1 **Figure 2** **Figure 3**

(a) Draw Figure 4 of the sequence. [2]

Figure 4 of the sequence is shown below.



(b) Study the pattern of numbers in each row of the table below. Each row relates to one of the figures in the sequence of figures **on page 23**. Some rows have not been included in the table.

Complete the rows numbered (i), (ii) and (iii).

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	Figure 1	Number of Outer Lines of Unit Length	Perimeter	
	1	$1 + 2 + 2$	5	
	2	$2 + 2 + 4$	8	
	3	$3 + 2 + 6$	11	
	⋮	⋮	⋮	
(i)	6	_____	_____	[2]
	⋮	⋮	⋮	
(ii)	_____	_____	65	[2]
	⋮	⋮	⋮	
(iii)	n	_____	_____	[2]

(i) For Figure 6,

$L = 6 + 2 + 2(6)$

$= 6 + 2 + 12$

For Figure 6,

Perimeter $= 6 + 2 + 12$

$= 20$

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(ii) Since the perimeter $= 65$

Then, $3n + 2 = 65$

$3n = 65 - 2$

$3n = 63$

$n = \frac{63}{3}$

$n = 21$

For Figure 21,

$L = 21 + 2 + 2(21)$

$= 21 + 2 + 42$

(iii) For Figure n,

$L = n + 2 + 2n$

Perimeter $= 3n + 2$

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(c) Show that no figure can have a perimeter of 100 units. [2]

If $P = 100$, then

$3n + 2 = 100$

$3n = 100 - 2$

$3n = 98$ (not divisible by 3)

$n = \frac{98}{3}$

However, n must be a positive integer and $\frac{98}{3}$ is not. So, $P \neq 100$ units.

Total: 10 marks

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SECTION II

Answer ALL questions.

ALGEBRA, RELATIONS, FUNCTIONS AND GRAPHS

8. (a) (i) Complete the table below for the function $f(x) = 3 + 2x - x^2$. [1]

x	-2	-1	0	1	2	3	4
$f(x)$	-5	0	3	4	3	0	-5

$f(-2) = 3 + 2(-2) - (-2)^2$

$= 3 - 4 - 4$

$= -5$

$f(2) = 3 + 2(2) - (2)^2$

$= 3 + 4 - 4$

$= 3$

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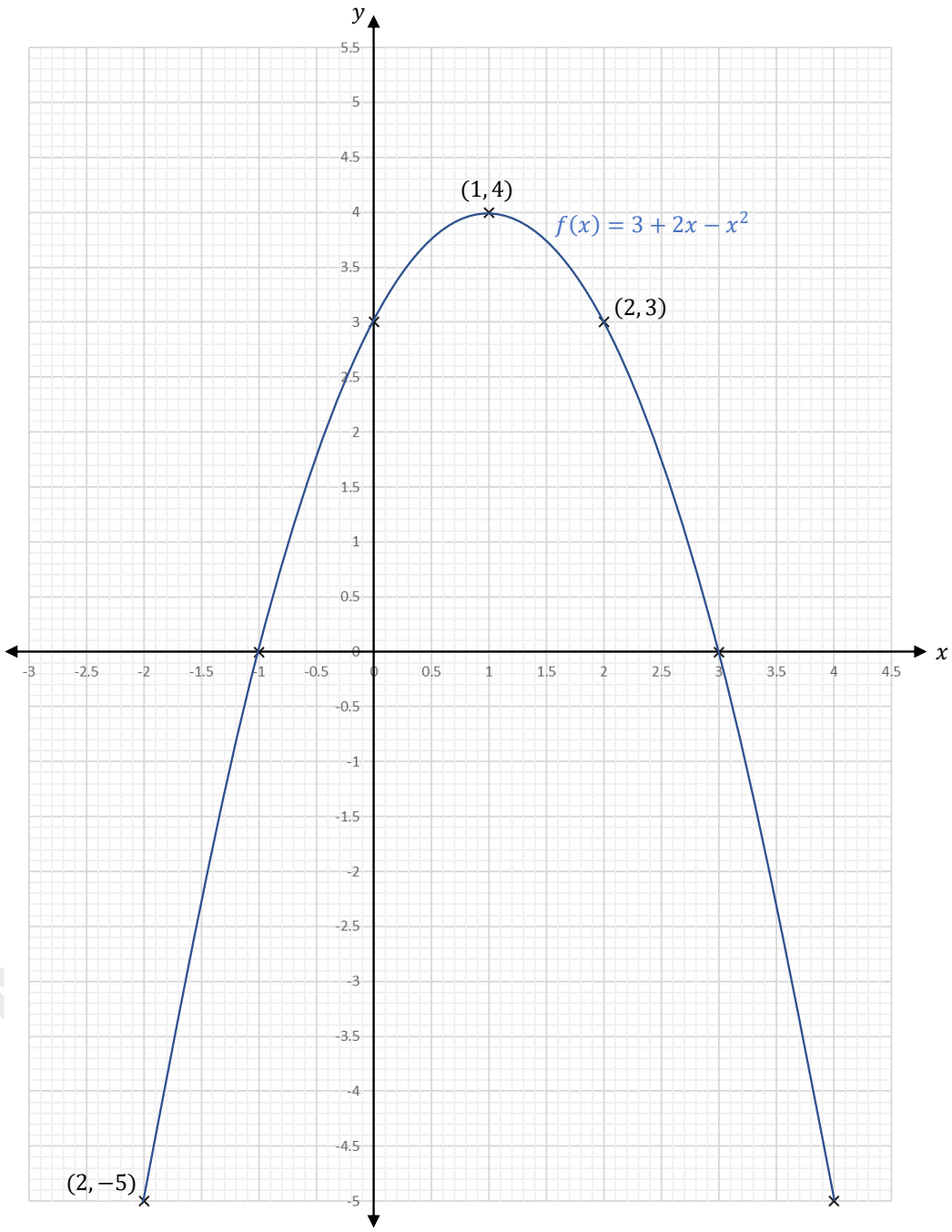
(ii) Complete the grid below to show all the points in the table on page 25 and hence, draw the graph of the function $f(x) = 3 + 2x - x^2$ for $-2 \leq x \leq 4$. [2]

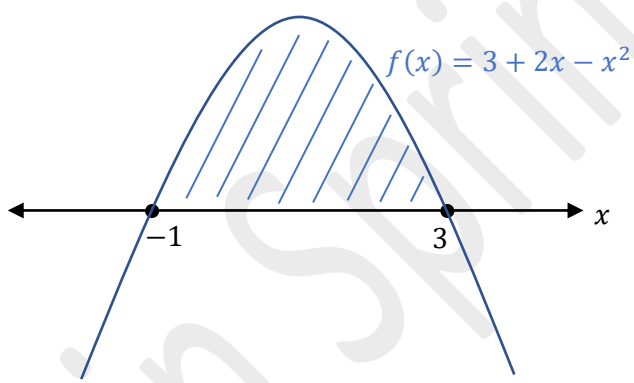


(iii) Using the graph **on page 26**, determine

(a) the coordinates of the maximum point of $f(x)$ [1]

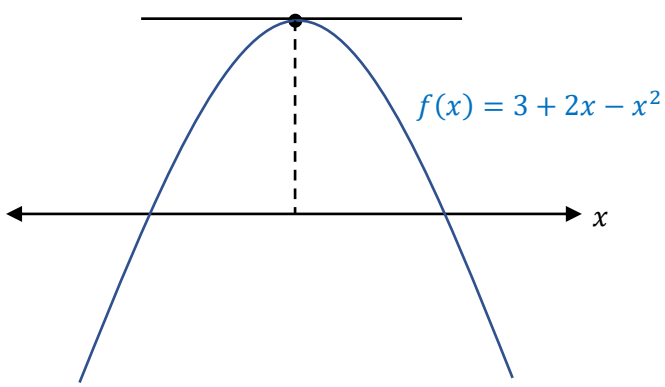
From the graph, the maximum turning point of $f(x)$ is $(1, 4)$.

(b) the range of values of x for which $f(x) > 0$ [2]



$\therefore$ The range of values of x for which $f(x) > 0$ is $-1 < x < 3$.

(c) the gradient of $f(x)$ at $x = 1$ [1]



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The tangent at $x = 1$ is a horizontal line whose gradient is zero.

Therefore, the gradient of $f(x)$ at $x = 1$ is 0.

(b) Mr. Thomas makes x bottles of juice and y cakes each day. To supply his customers, he makes at least 20 bottles of juice and no more than 15 cakes each day.

(i) Write TWO inequalities to represent this information. [2]

The number of bottles of juice is at least 20.

The inequality is: $x \geq 20$

The number of cakes is no more than 15.

The inequality is: $y \leq 15$

(ii) Each day, Mr. Thomas uses \$163 to make the bottles of juice and the cakes. The cost to make a bottle of juice is \$3.50 while the cost to make a cake is \$5.25.

Write an inequality to represent this information. [2]

The cost to make one bottle of juice is \$3.50.

The cost to make one cake is \$5.25.

Mr. Thomas spends \$163 each day to make the bottles of juice and the cakes.

The inequality is: $3.5x + 5.25y \leq 163$

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(iii) Show that on any given day, it is NOT possible for Mr. Thomas to make 50 bottles of juice and 12 cakes. [1]

50 bottles of juice at \$3.50 each $= 50 \times \$3.50$

$= \$175$

12 cakes at \$5.25 each $= 12 \times \$5.25$

$= \$63$

The total cost $= \$175 + \$63$

$= \$238$

Since \$238 is greater than \$163, it is not possible on his budget.

Total: 12 marks

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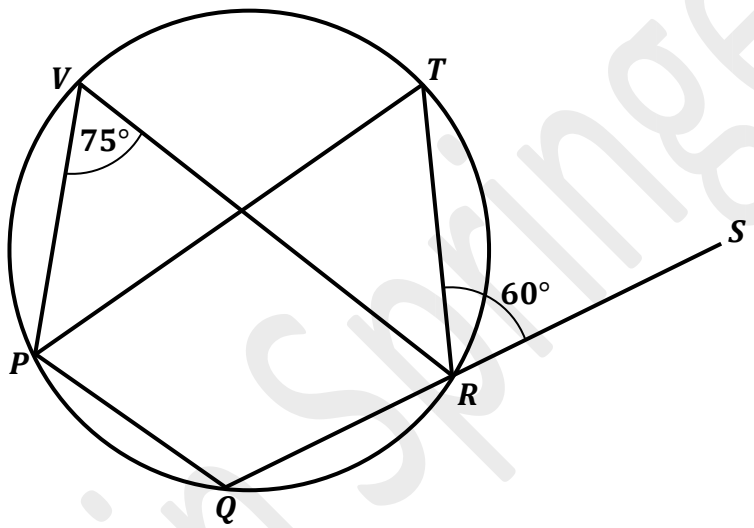
9. (a) The diagram below, **not drawn to scale**, shows a circle. The points P, Q, R, T and V are on the circumference. QRS is a straight line. Angle $PVR = 75°$ and angle $TRS = 60°$.



Determine the value of EACH of the following angles. Show detailed working where necessary and give a reason to support your answer.

(i) Angle PTR [2]

Consider the chord PR.

The angles subtended by a chord (PR) at the circumference of a circle $(P\hat{V}R$ and $P\hat{T}R)$ and standing on the same arc are equal.

Hence, $P\hat{T}R = 75°$.

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(ii) Angle TPQ [2]

Angles in a straight line add up to 180°.

$T\hat{R}Q = 180° - 60°$

$= 120°$

The opposite angles, $T\hat{P}Q$ and $T\hat{R}Q$, of a cyclic quadrilateral are supplementary.

$T\hat{P}Q = 180° - 120°$

$= 60°$

(iii) Obtuse angle POR where O is the centre of the circle. [2]

The angle subtended by a chord (PR) at the center of a circle $(P\hat{O}R)$ is twice the angle that the chord subtends at the circumference $(P\hat{T}R)$ standing on the same arc.

$\therefore P\hat{O}R = 2(75°)$

$= 150°$

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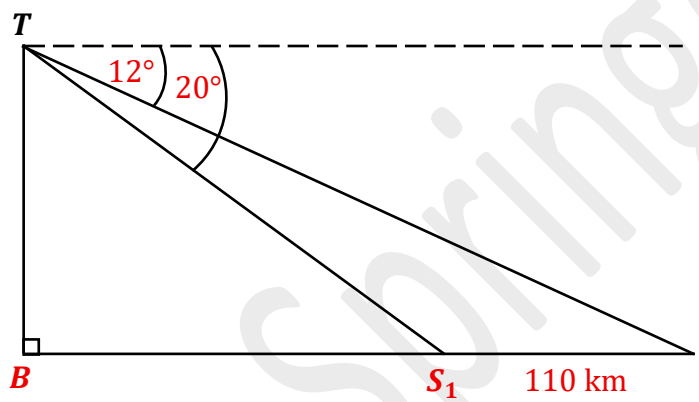
(b) A person at the top of a lighthouse, TB, sees two ships, S_1 and S_2, approaching the coast as illustrated in the diagram below. The angles of depression are 12° and 20° respectively. The ships are 110 m apart.

(i) Complete the diagram below by inserting the angles of depression and the distance between the ships. [1]



(ii) Determine, to the nearest metre,

(a) the distance, TS_2, between the top of the lighthouse and Ship 2 [3]

Consider ΔTS_1S_2.

Angle $S_1TS_2 = 20° - 12°$

$= 8°$

Angle $BTS_1 = 90° - 20°$

$= 70°$

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Angle $TS_2B = 12°$ (alternate angles)

All angles in a triangle add up to 180°.

Angle $TS_1S_2 = 180° - (8° + 12°)$

$= 180° - 20°$

$= 160°$

Using the sine rule,

$\frac{TS_2}{\sin 160°} = \frac{110}{\sin 8°}$

$TS_2 = \frac{110 \times \sin 160°}{\sin 8°}$

$= 270.3$

$= 270\ m$ (to the nearest metre)

(b) the height of the lighthouse, TB [2]

Consider ΔTBS_2.

$\sin BS_2T = \frac{TB}{TS_2}$

$\sin 12° = \frac{TB}{270.3}$

$TB = \sin 12° \times 270.3$

$TB = 56\ m$ (to the nearest metre)

Total: 12 marks

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10. (a) Three matrices are given as follows:

$$P = \begin{pmatrix} -1 & 2 \\ 0 & 5 \end{pmatrix}, Q = \begin{pmatrix} a \\ b \end{pmatrix} \text{ and } R = \begin{pmatrix} 11 \\ 15 \end{pmatrix}.$$

(i) Using a calculation to support your answer, explain whether matrix P is a singular or a non-singular matrix. [2]

We are given that $P = \begin{pmatrix} -1 & 2 \\ 0 & 5 \end{pmatrix}$.

$\det P = ad - bc$

$= (-1)(5) - (2)(0)$

$= -5 - 0$

$= -5$

Since the $\det P \neq 0$, then P is a non-singular matrix.

(ii) Given that $PQ = R$, determine the values of a and b. [3]

$PQ = R$

$$\begin{pmatrix} -1 & 2 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 11 \\ 15 \end{pmatrix}$$

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$$\begin{pmatrix} (-1 \times a) + (2 \times b) \\ (0 \times a) + (5 \times b) \end{pmatrix} = \begin{pmatrix} 11 \\ 15 \end{pmatrix}$$

$$\begin{pmatrix} -a + 2b \\ 5b \end{pmatrix} = \begin{pmatrix} 11 \\ 15 \end{pmatrix}$$

Comparing the equivalent matrices and equating corresponding entries, we get:

$5b = 15$

$b = \frac{15}{5}$

$b = 3$

Now,

$-a + 2b = 11$

$-a + 2(3) = 11$

$-a + 6 = 11$

$-a = 11 - 6$

$-a = 5$

$a = -5$

$\therefore a = -5$ and $b = 3$

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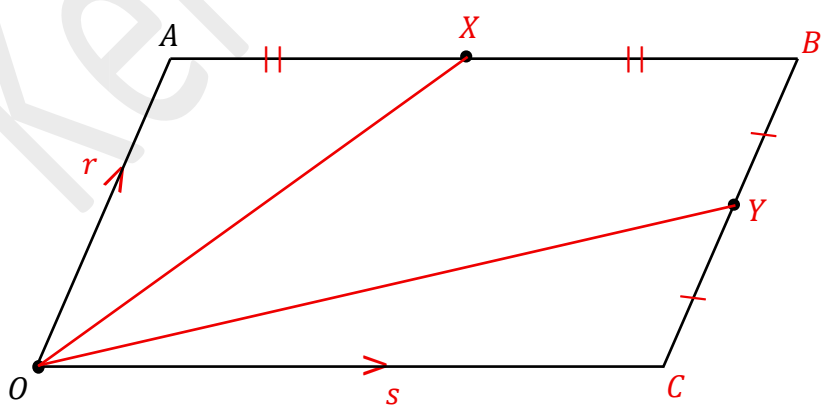
(iii) State the reason why the matrix product QP is not possible. [1]

The matrix multiplication QP is not possible because the number of columns in Q is not equal to the number of rows in P.

(b) $OABC$ is a parallelogram. $X$is the midpoint of AB and Y is the midpoint of BC.

$\overrightarrow{OA} = \boldsymbol{r}$ and $\overrightarrow{OC} = \boldsymbol{s}$.

(i) Complete the diagram below to represent ALL the information given above. [3]



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(ii) Given that $\overrightarrow{OX} + \overrightarrow{OY} = k(\boldsymbol{r} + \boldsymbol{s})$, where k is a constant, using a vector method, find the value of k. [3]

Since $\overrightarrow{AB}$ and $\overrightarrow{OC}$ are opposite sides of the parallelogram, then

$\overrightarrow{AB} = \overrightarrow{OC}$

$= s$

Since X is the midpoint of AB, then

$\overrightarrow{AX} = \frac{1}{2}\overrightarrow{AB}$

$= \frac{1}{2}s$

Now,

$\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AX}$

$= r + \frac{1}{2}s$

Since $\overrightarrow{CB}$ and $\overrightarrow{OA}$ are opposite sides of the parallelogram, then

$\overrightarrow{CB} = \overrightarrow{OA}$

$= r$

Now,

$\overrightarrow{OY} = \overrightarrow{OC} + \overrightarrow{CY}$

$= s + \frac{1}{2}r$

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$$\therefore \overrightarrow{OX} + \overrightarrow{OY} = r + \frac{1}{2}s + s + \frac{1}{2}r$$

$$= \frac{3}{2}r + \frac{3}{2}s$$

$$= \frac{3}{2}(r + s) \quad \text{which is of the form } k(r + s)\text{, where } k = \frac{3}{2}.$$

Total: 12 marks

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.